

4	(a)	(i)1	<u>distance moved by wavefront/wave during one period / during one oscillation of a particle in that wave</u> OR <u>minimum distance between two wavefronts/crests/troughs/peaks</u> OR <u>minimum distance between two points which are in phase</u>	B1 (B1) (B1)
		(i)2	The phase difference between two particles refers to <u>how much one particle lags or leads another with respect to a cycle.</u> OR <u>how one wave lags or leads another wave of the same wavelength.</u>	B1 (B1)
		(ii)	$\frac{\Delta\theta}{2\pi} = \frac{\Delta x}{\lambda}$ OR $\frac{\Delta\theta}{360^\circ} = \frac{\Delta x}{\lambda}$	B1
	(b)	(i)	Using $v = \frac{\lambda}{T}$ or $v = f\lambda$ and $f = \frac{1}{T}$ From Fig. 4.1, $T = 0.60$ s $\lambda = vT = (20)(0.60)$ $= 12$ cm	M1 A1
	(b)	(i)	$\frac{\Delta\theta}{360^\circ} = \frac{\Delta t}{T} = \frac{0.20}{0.60}$ $\Delta\theta = 120^\circ$ (or $360^\circ - 120^\circ = 240^\circ$) (Award M1 mark only if answers expressed in radians (2.09 rad).)	M1 A1

		(ii)	At $t = 0.45$ s, using the principle of superposition, net displacement $= 1.00 + (-3.00)$ $= -2.00$ mm (Accept ± 0.1 mm)	M1 A1
		(iv)	Intensity $\propto$ (amplitude) ² $\frac{I_Q}{I_P} = \left(\frac{A_Q}{A_P} \right)^2 = \left(\frac{2.0}{3.0} \right)^2$ $= 0.444 \text{ (3sf)}$	M1 A1
		(v)	Two sources (waves) are said to be coherent when <u>the phase difference is always the same</u> OR there is a <u>constant phase relationship</u> at all times.	B1

5	(c)	Current $= 2.7 - 1.5 = 1.2$ A	B1
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